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(Ministry of Education Initiative)

REASEARCH ORIENTED REAL TIME PROJECT - 2026

WEPON DETECTION USING FASTER RCNN

Guide :

Dr . J. SWARNALATHA
Assistant Professor

Co- ordinator:

M.SRILEKHA
DR.B.VENKATESWARULU
NAYAK
A.DIVYABHARATHI

Presenting By:-

Batch 12

🎓 **K.NITHIN – 247 R1A05BD**

🎓 **N.SAILAJA – 247 R1A05BR**

🎓 **N.SRAVANTHI- 247 R1A05BU**

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Abstract

Security is always a main concern in every domain, due to a rise in crime rate in a crowded event or suspicious lonely areas.

Abnormal detection and monitoring have major applications of computer vision to tackle various problems.

Due to growing demand in the protection of safety, security and personal properties, needs and deployment of video surveillance systems can recognize and interpret the scene and anomaly events play a vital role in intelligence monitoring.

This paper implements automatic gun (or) weapon detection **using a Faster RCNN algorithms**.

One dataset, which had pre-labelled images and the other one is a set of images, which were labelled manually.

Results are tabulated, both algorithms achieve **good accuracy**, but their application in real situations can be based on the trade-off between speed and accuracy.

Existing System and Disadvantages

Existing System:-

Abnormal detection and monitoring have major applications of computer vision to tackle various problems. Due to growing demand in the protection of **safety, security** and personal properties, needs and deployment of video surveillance systems can recognize and interpret the scene and anomaly events play a vital role in intelligence monitoring. This paper implements **automatic gun (or) weapon detection** using a convolution neural network (CNN)

Disadvantages:-

- Fast RCNN is slower than FRCNN.
- It takes more time to clear.

Proposed System and Advantages

Proposed System:-

In this project we have developed a Faster R-CNN (FRCNN) based Weapon Detection System where the user can upload a **weapon image dataset** (images with XML bounding box annotations) and train the detection model from scratch.

The system **reads all annotated XML files**, extracts bounding box coordinates, and trains the FRCNN model for accurate weapon localization and classification.

After training, the model gets saved and can be used to **detect weapons** from both **images and videos**.

The user can upload a test image to detect weapons with bounding boxes and confidence scores.

Similarly, the user can upload an **MP4 video file**, and the system performs frame-by-frame weapon detection. The system also generates an **Accuracy–Loss graph** to monitor model performance during training.

All detection results are displayed visually with bounding boxes drawn around detected weapons.

Advantages:-

- More Accuracy.
- FRCNN is faster than CNN



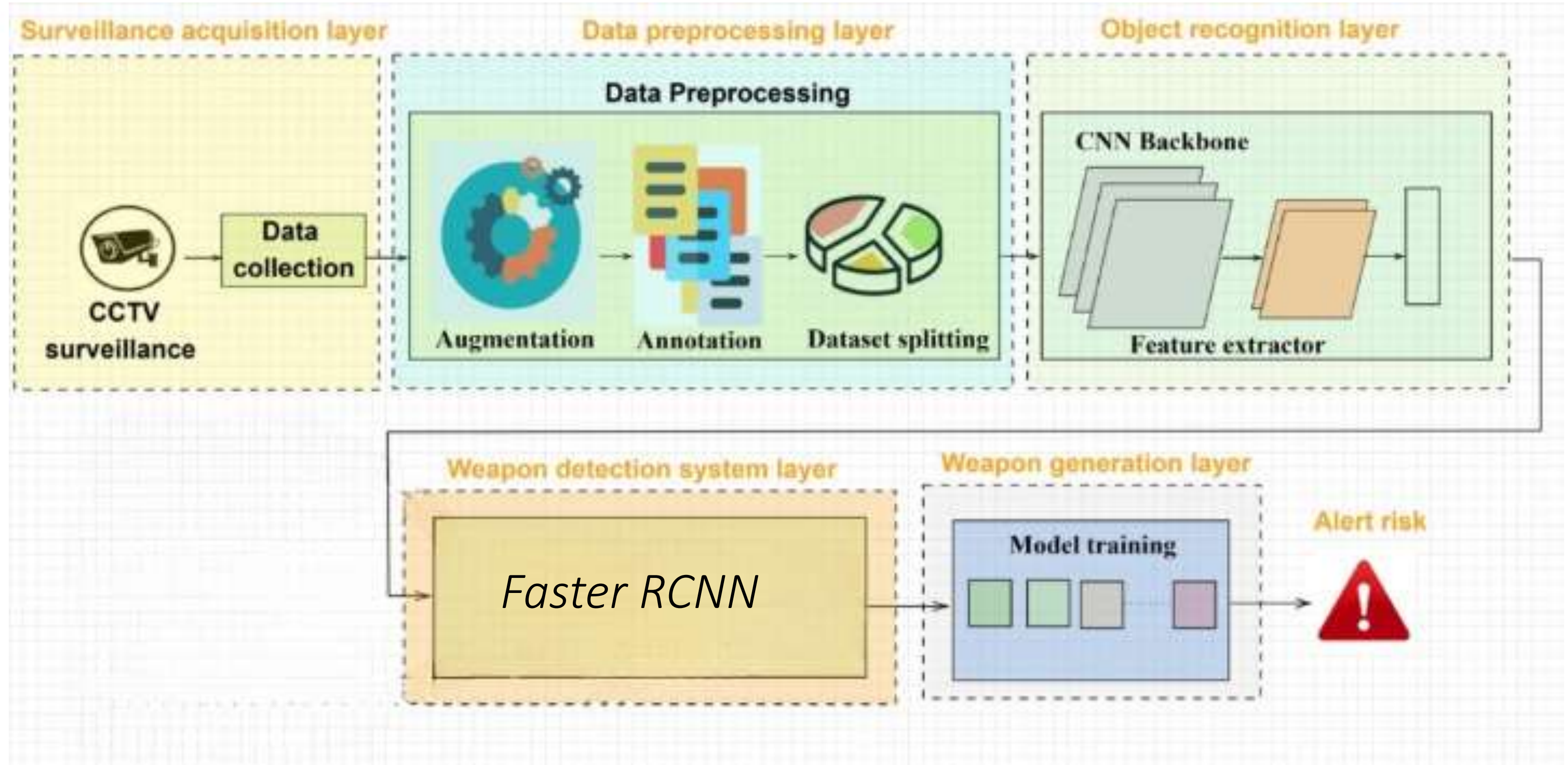
Hardware Requirements

Processor	- Intel i3 or i4
Speed	- 1.1 GHz
RAM	- 4 GB (min)
Hard Disk	- 500 GB (min)
Key Board	- Standard Windows Keyboard
Mouse	- Two or Three Button Mouse
Monitor	- SVGA

Software Requirements

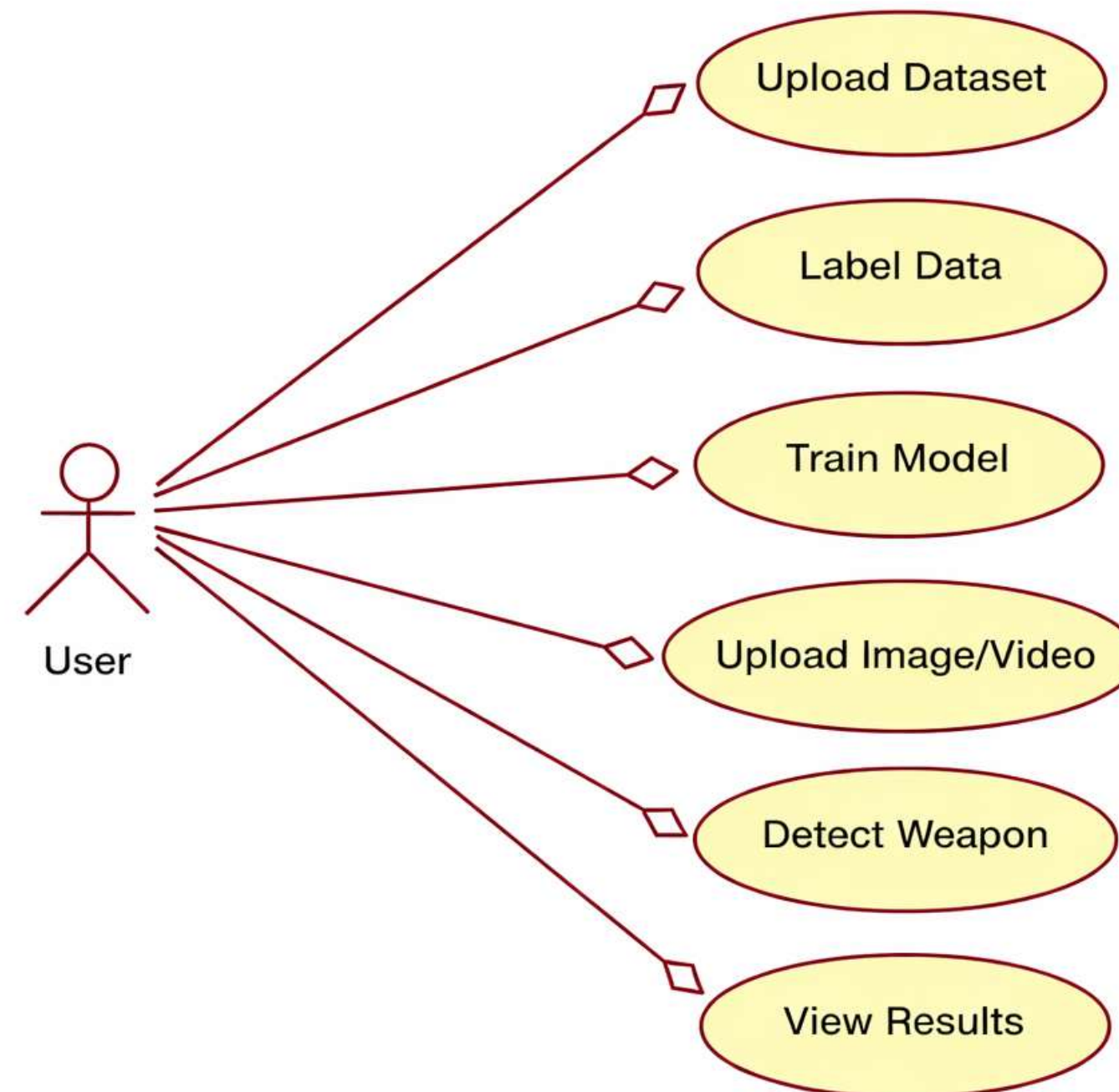
Operating System	- Windows 10 or above
Programming Language	- Python(3.7.0)

System Architecture



Use case Diagram

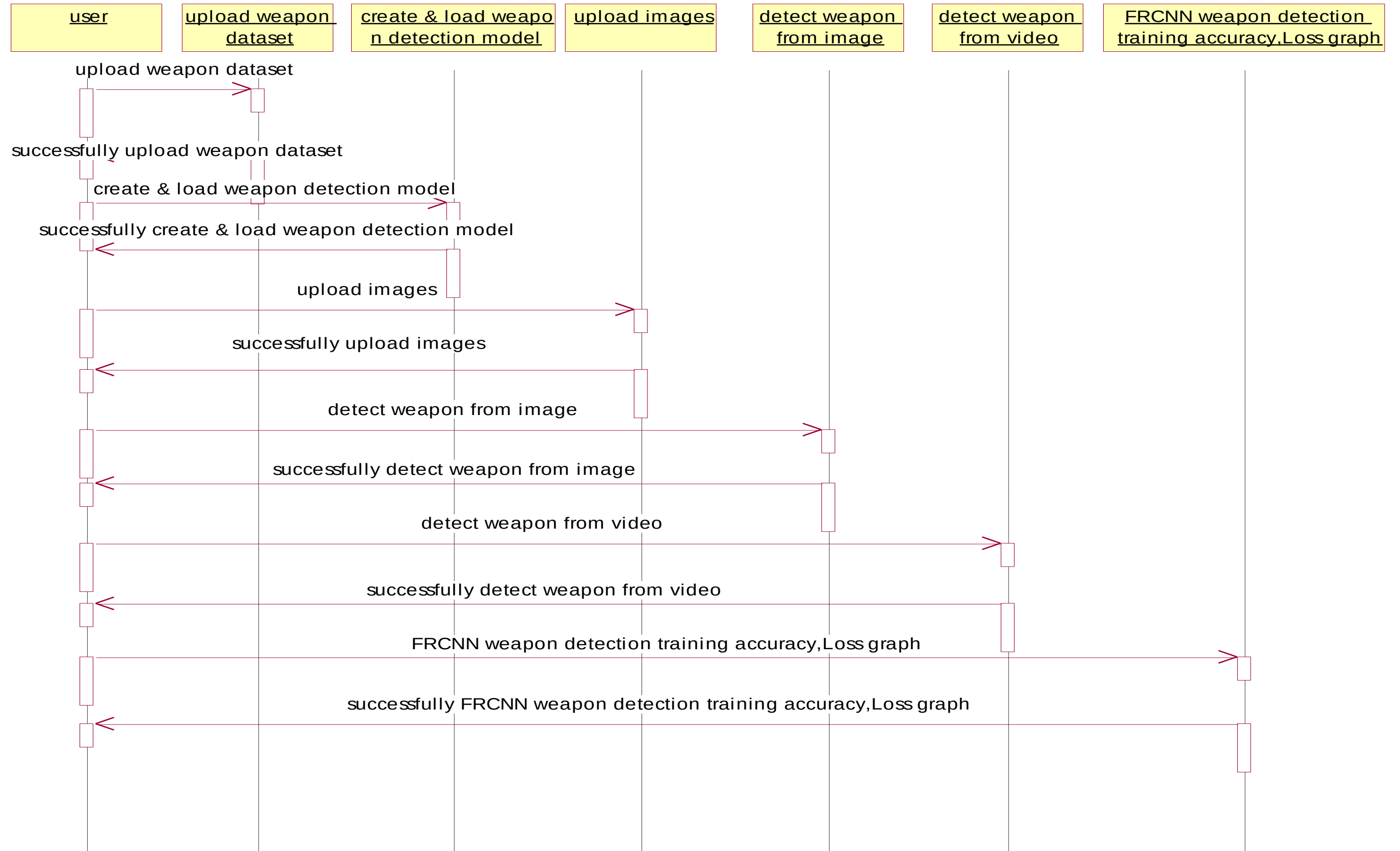
This use case diagram represents the interaction between the user and the weapon detection system. The user can perform multiple actions such as uploading datasets, training the model, testing images or videos, and viewing detection results.



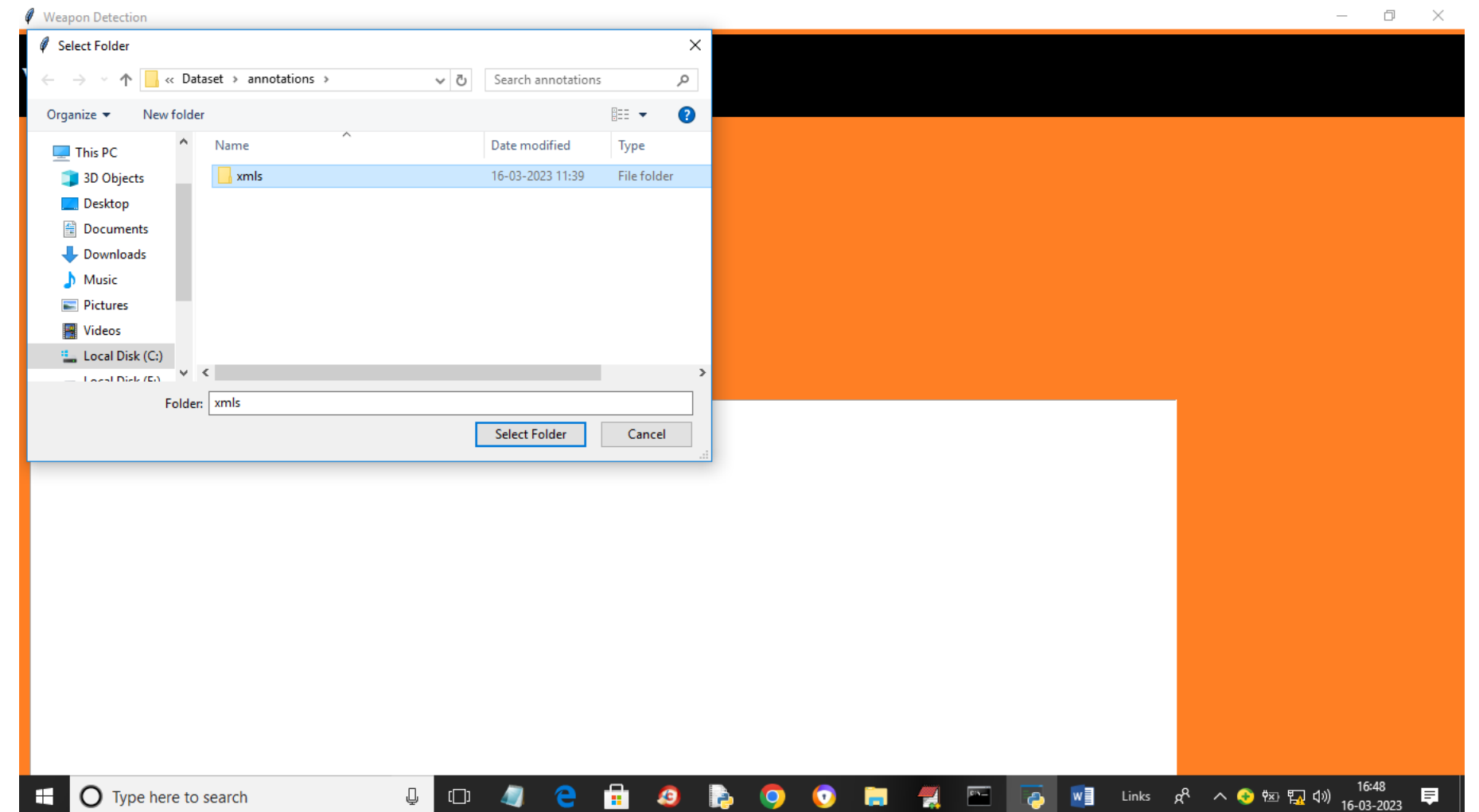
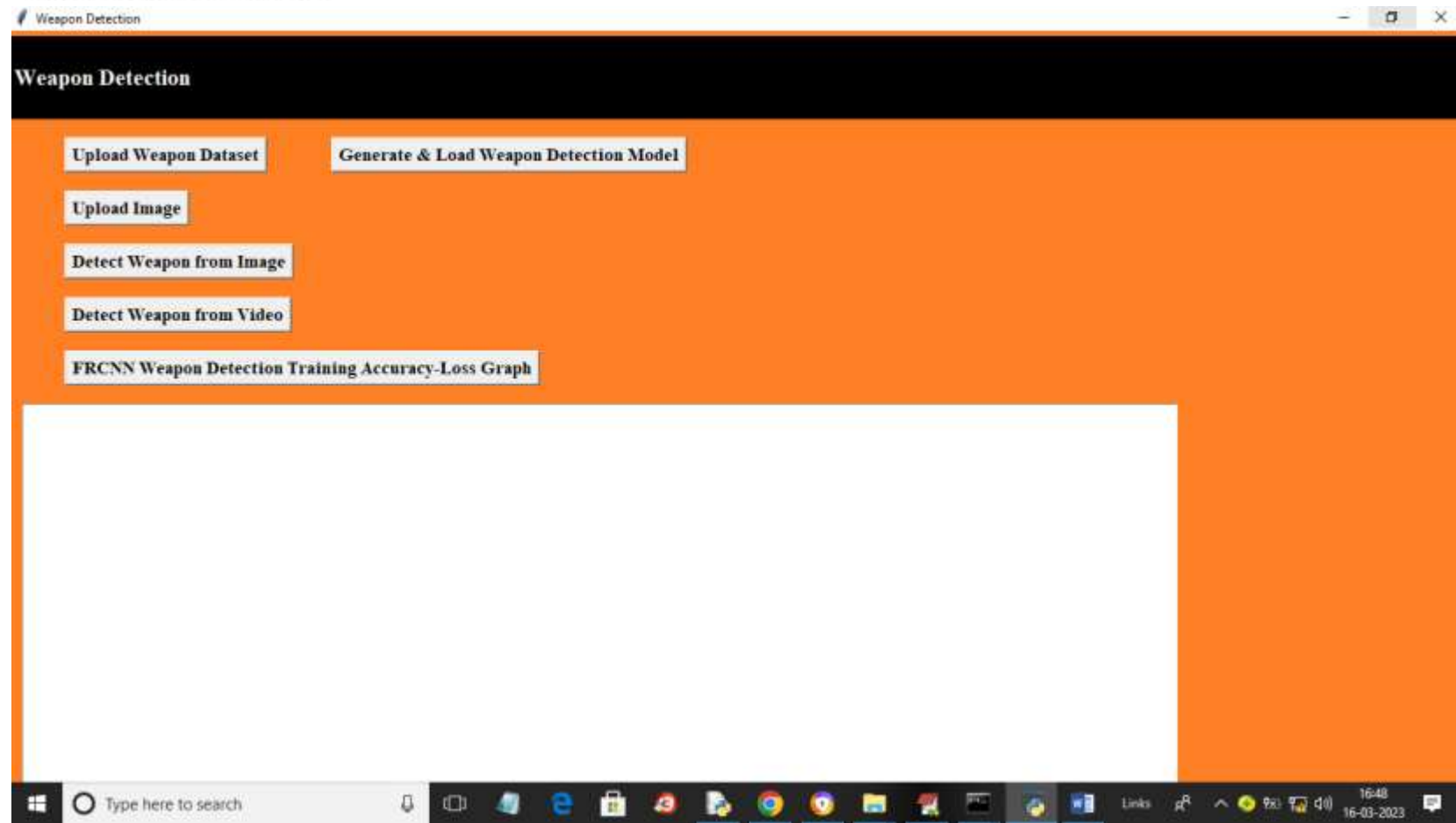
Class Diagram



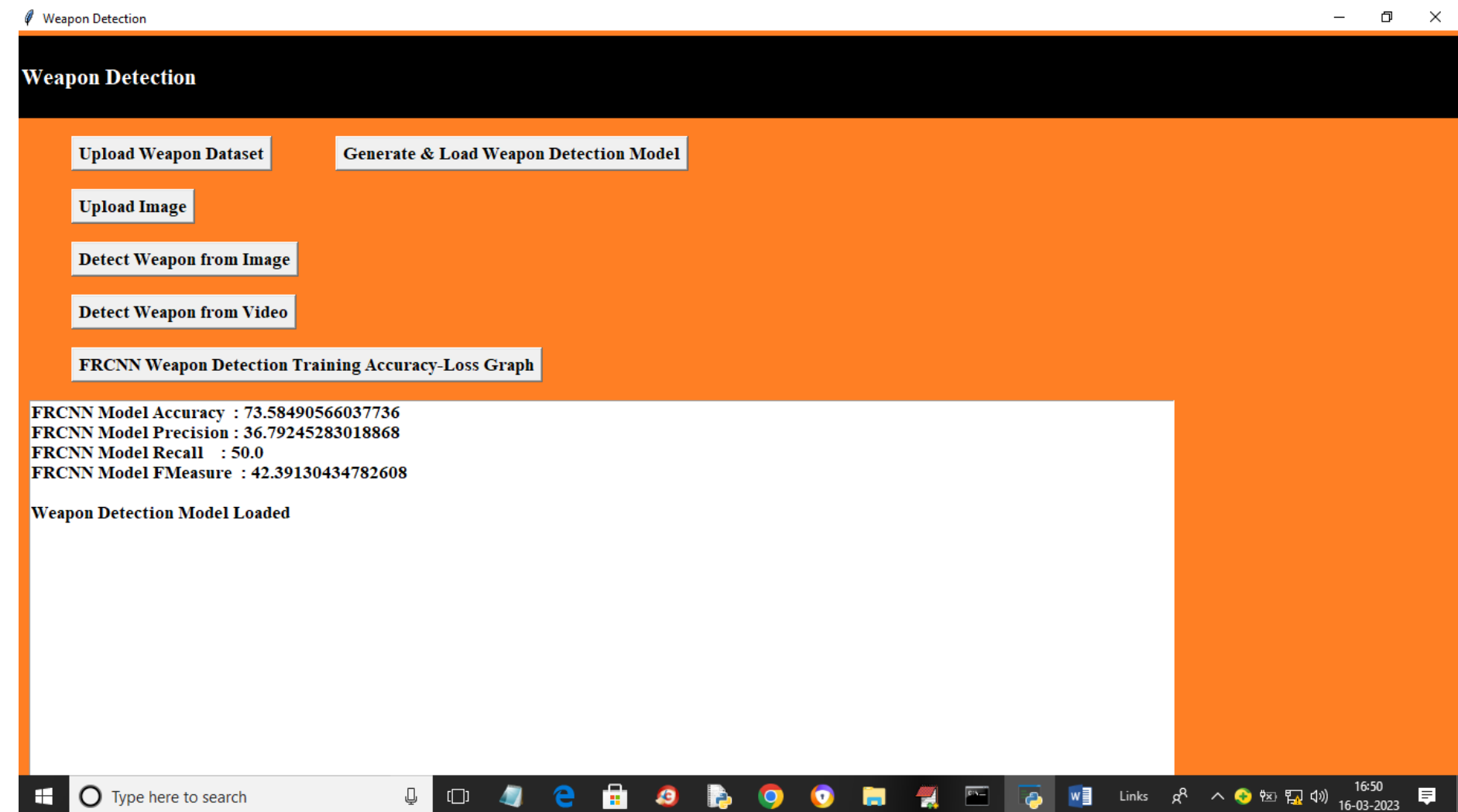
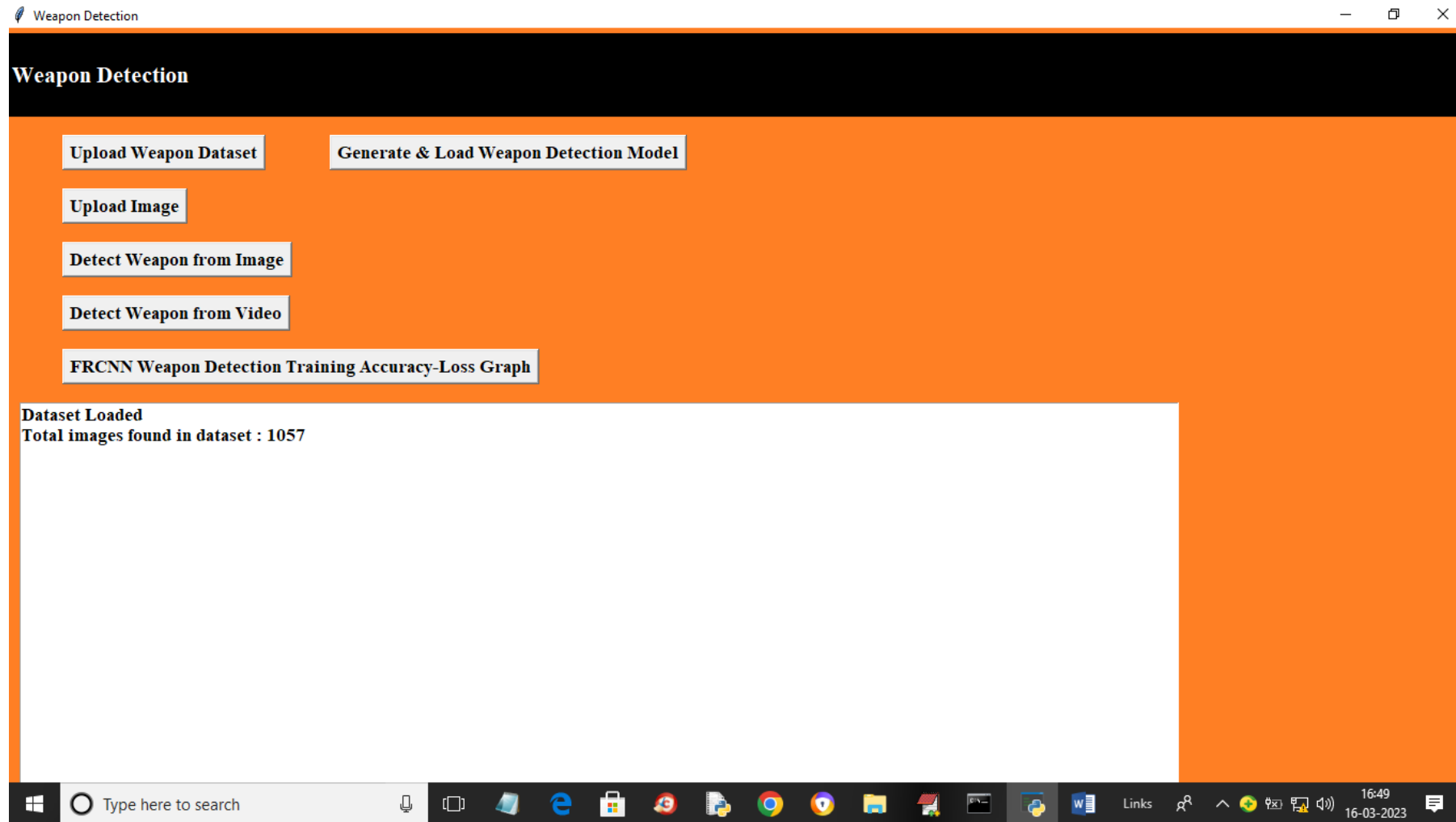
Sequence Diagram



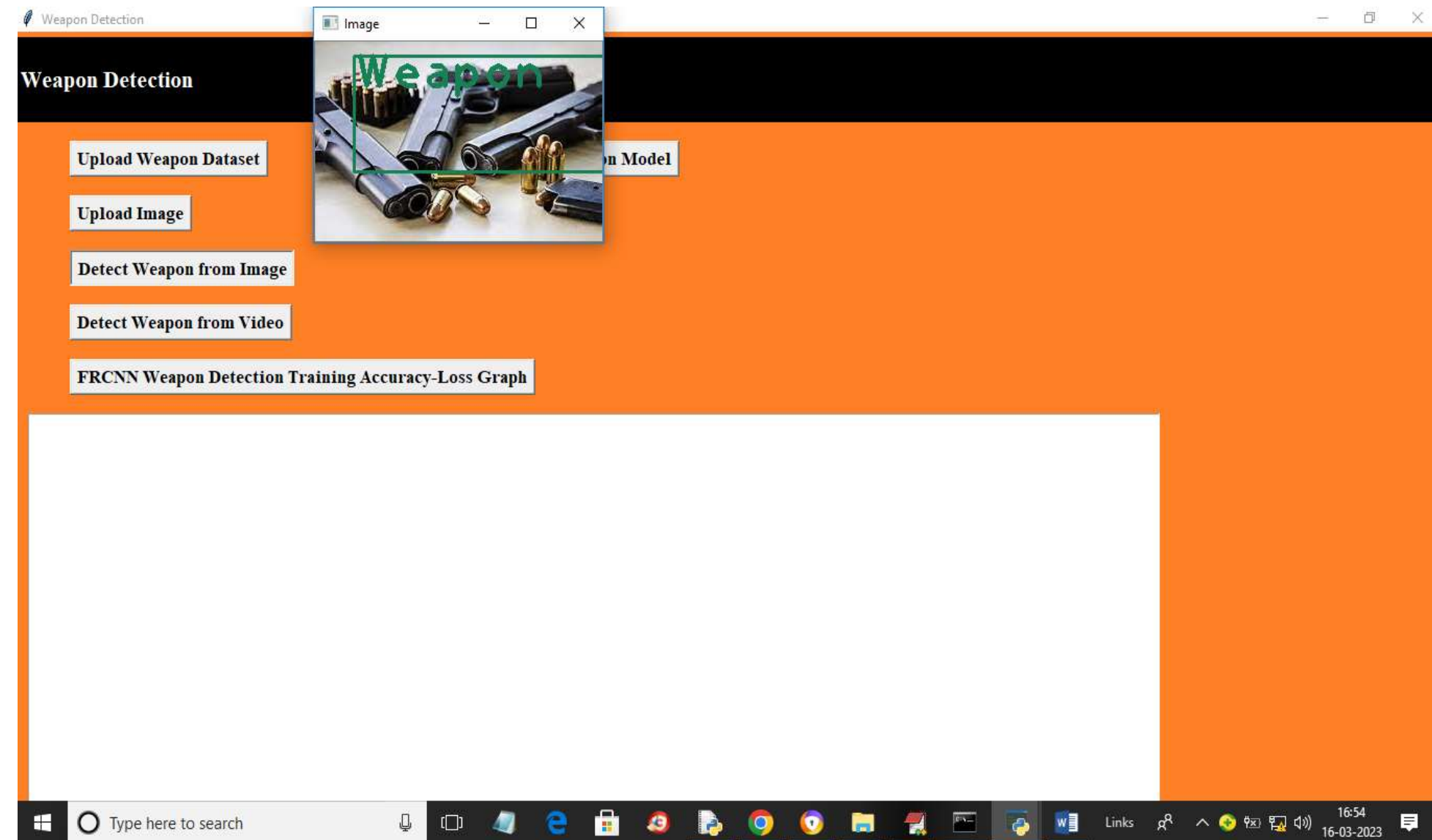
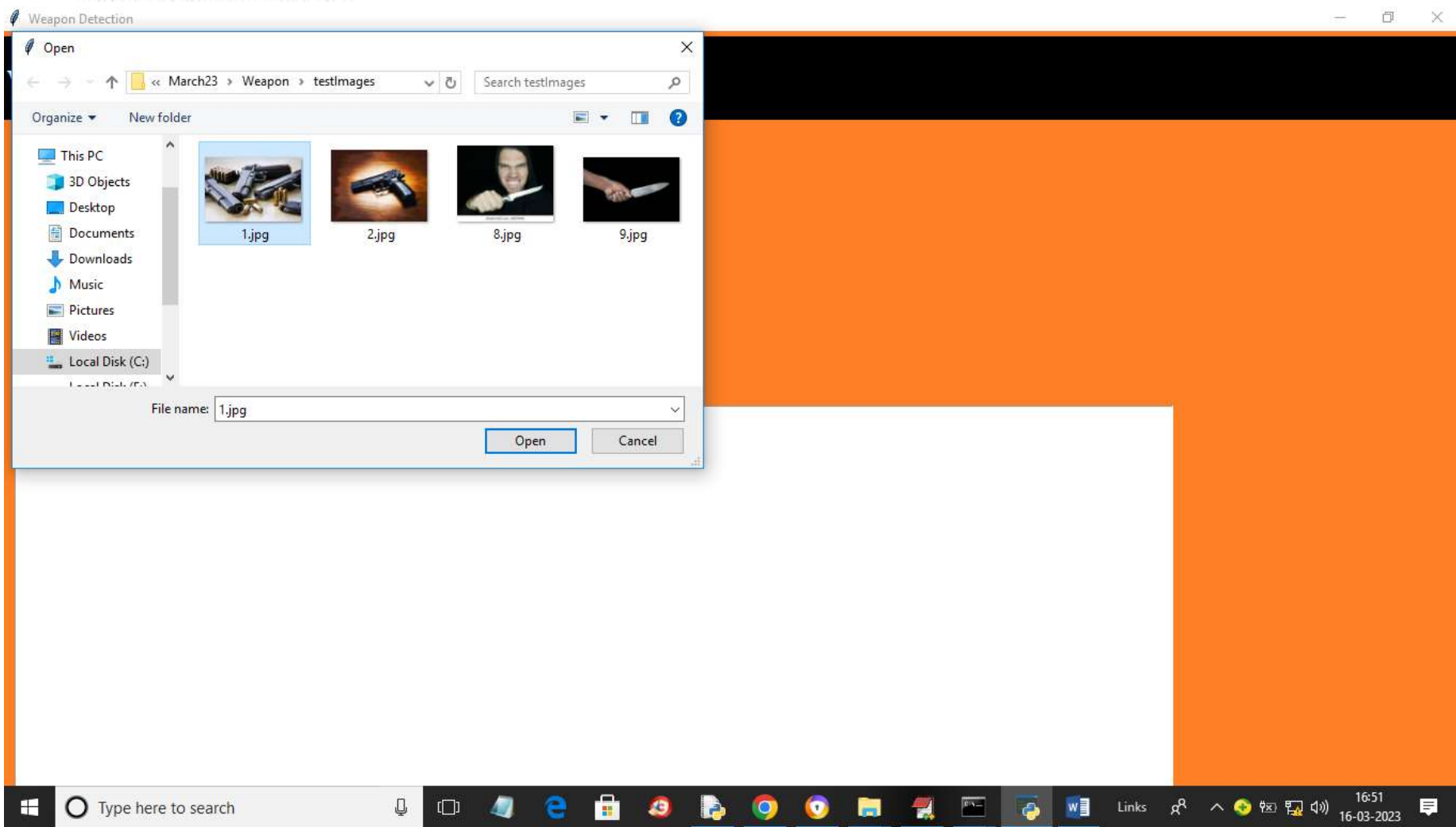
UI Design



UI Design

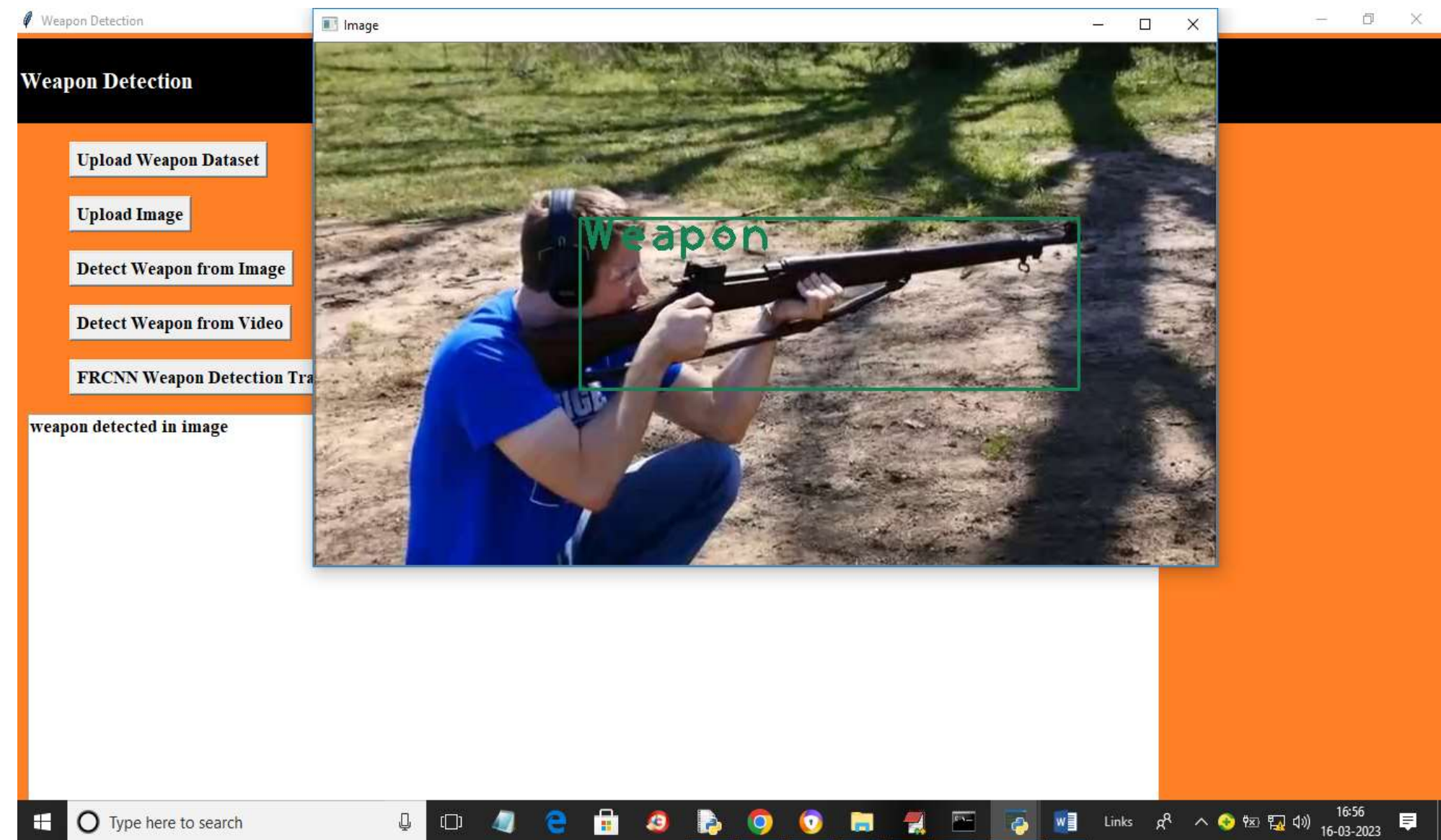
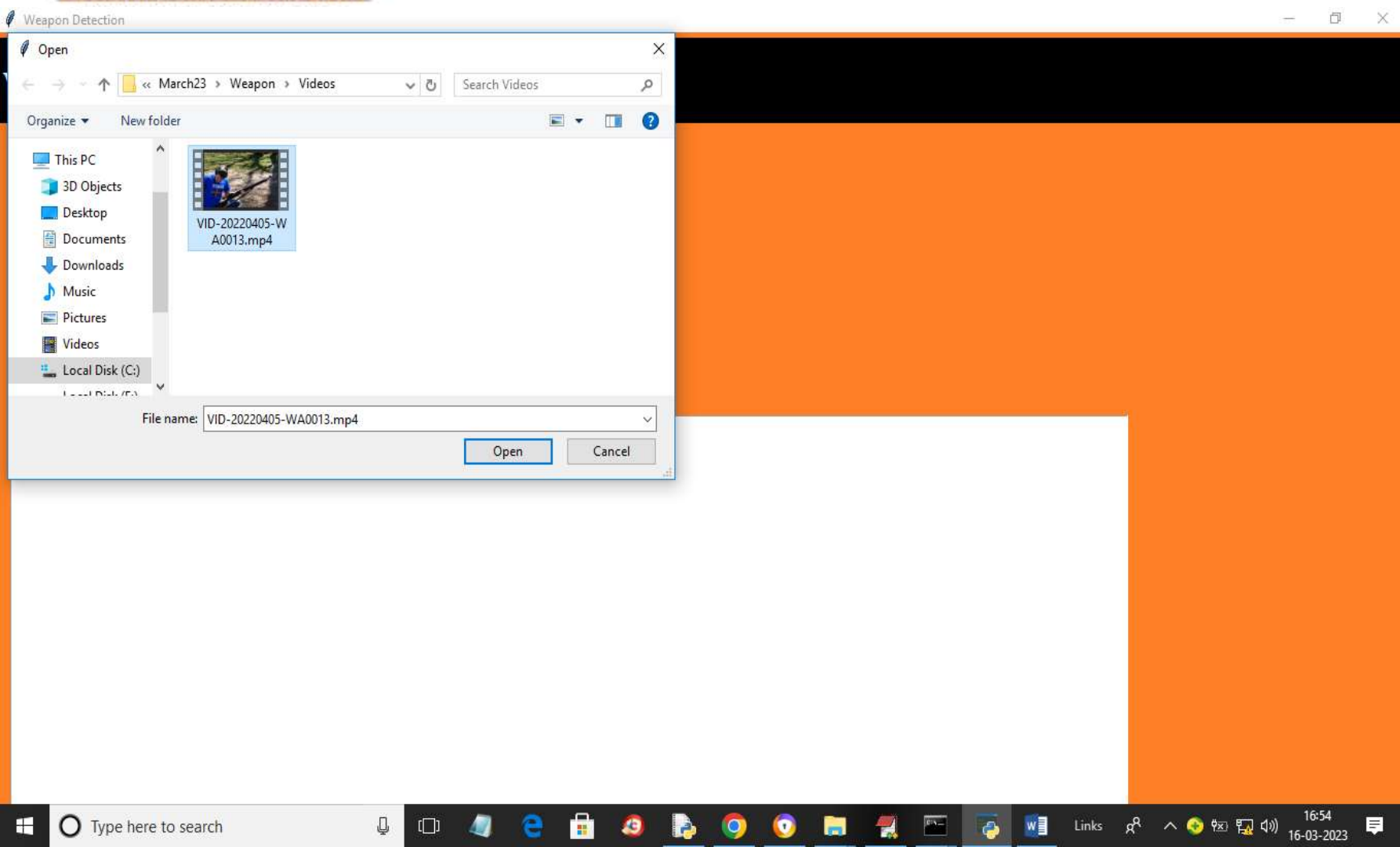


UI Design

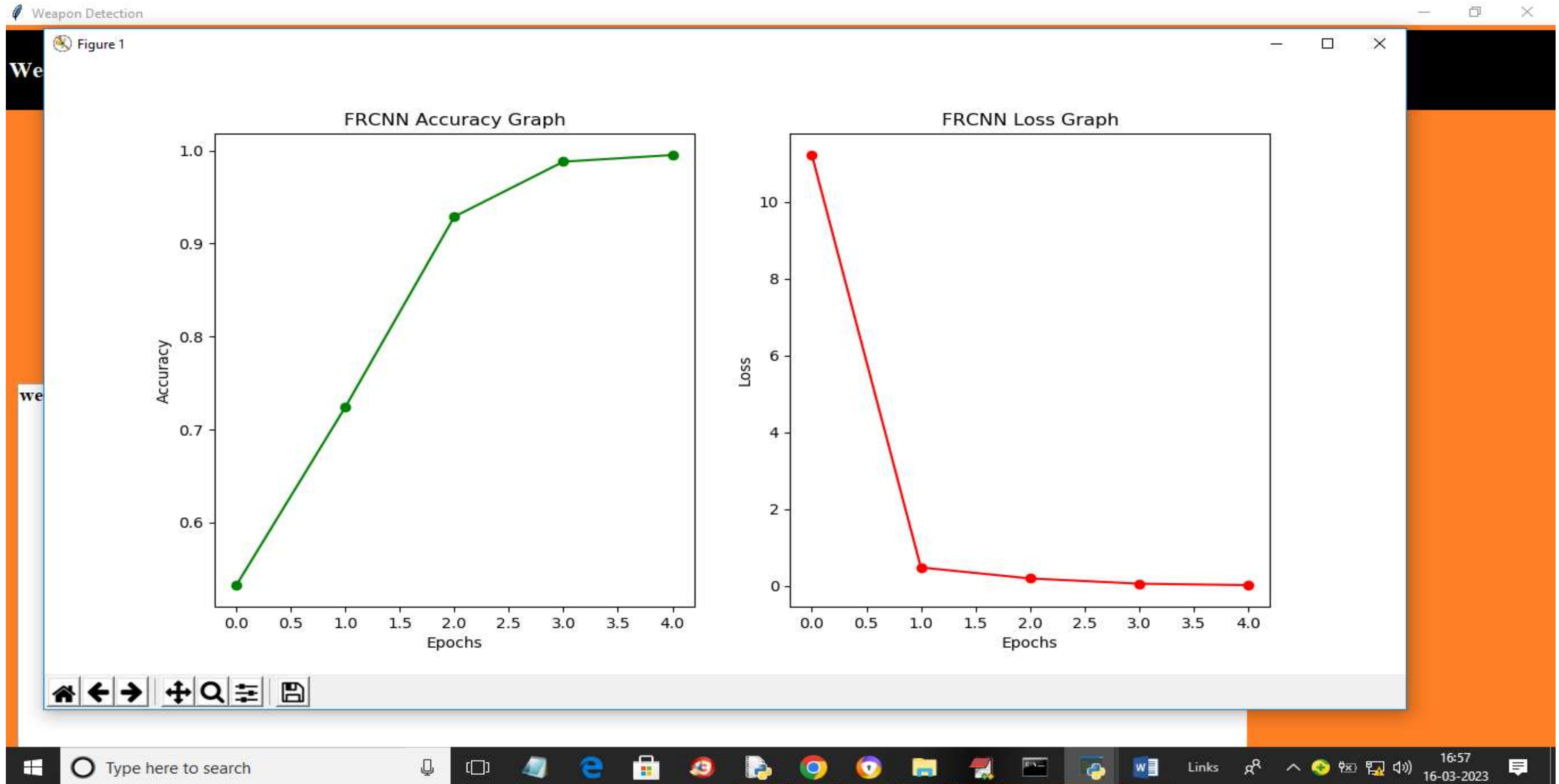




UI Design



UI Design



NOVELTY OF YOUR PROJECT

- Automatic **weapon detection** in **surveillance images** or **videos**
- Uses **deep learning** for **higher accuracy**
- Can detect **multiple weapon types** (gun, knife, etc.)
- Helps improve **public safety** by **early threat detection**

MODULES

Data Collection: Collect images of weapons like guns and knives.

Data Preprocessing: Prepare and label the images using Labeling.

Model Training: Train the detection model using Faster R-CNN.

Weapon Detection: The system identifies weapons in images or video frames.

Result Display: The detected weapon is shown with a bounding box on the screen.

Conclusion

- SSD and Faster RCNN algorithms are simulated for pre labeled and self-created image dataset for weapon (gun) detection. Both the algorithms are efficient and give good results but their application in real time is based on a tradeoff between speed and accuracy. In terms of speed, SSD algorithm gives better speed with 0.736 s/frame.
- Whereas Faster RCNN gives speed 1.606s/frame, which is poor compared to SSD. With respect to accuracy, Faster RCNN gives better accuracy of 84.6%. Whereas SSD gives an accuracy of 73.8%, which is poor compared to faster RCNN.
- SSD provided real time detection due to faster speed but Faster RCNN provided superior accuracy. Further, it can be implemented for larger datasets by training using GPUs and high-end DSP and FPGA kits.

References

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THANK YOU